KYLE DARDEN

Software Engineer | Data & Backend Systems

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Software engineer with a backend and data-systems focus, building services end to end — FastAPI APIs over PostgreSQL, ingestion pipelines, and dbt modeling — with CI and automated tests throughout. Physics-trained, with quantitative depth; strong in Python and system design. Currently backfilling CS fundamentals toward a master's in computer science.

SKILLS

Languages: Python, SQL, TypeScript

Backend & APIs: FastAPI, REST, SQLAlchemy, Alembic, AWS Cognito

Data: PostgreSQL, dbt, dimensional modeling, ETL/ELT, data warehousing

Infrastructure: Docker, GitHub Actions (CI/CD), Render, Git

Testing: pytest, Playwright

EXPERIENCE

Co-Founder & Backend Developer — FreightFolio Mar 2025 – Dec 2025

Logistics SaaS · Remote

- Architected the backend as five modular FastAPI services over PostgreSQL, each with Alembic migrations and containerized with Docker.
- Built a production-grade authentication layer on AWS Cognito — JWKS fetching with a thread-safe cache, key rotation, and RS256 signature/audience/issuer verification — enforced at the router level across all services.

Server — Fairmont Austin (Garrison) Oct 2024 – Present

Austin, TX

- Fine-dining service role held through a career change into software and ongoing technical coursework.

PROJECTS

CS2 Analytics — live, end-to-end data platform

github.com/dardenkyle/CS2-analytics · dardenkyle.github.io/CS2-analytics

- Built the full pipeline — Python ingestion, a Render-hosted FastAPI service and PostgreSQL database, and a React/TypeScript dashboard on GitHub Pages — deployed end to end.
- Modeled the analytics layer in dbt — staging, intermediate, and marts (fact and dimension tables) — including an SCD2 snapshot tracking player-team roster history, using a check-strategy snapshot chosen over a timestamp strategy to avoid spurious intervals from the source's lack of a reliable roster-change timestamp.
- Operable via a Typer CLI with incremental and backfill ingestion modes for running and reprocessing the pipeline.
- 137 automated tests with CI gates on both backend and frontend.

movie_analytics_etl — dimensional data warehouse

github.com/dardenkyle/movie_analytics_etl

- Kimball star-schema warehouse over 176M+ IMDb records built with dbt: staging and marts layers and ~30 data-quality tests, on PostgreSQL.

site-sentry — uptime & performance monitoring

github.com/dardenkyle/site-sentry · dardenkyle.github.io/site-sentry

- 64 Playwright/pytest checks against kyledarden.com, run twice daily via GitHub Actions, publishing live uptime and page-load latency (TTFB / DOM / load) trends from persisted run history.

EDUCATION

Associate of Science, Computer Science — Austin Community College 2026 – Present

Data Structures & Algorithms, Discrete Mathematics, Programming Fundamentals

Post-Graduate Certificate, Artificial Intelligence & Machine Learning — Texas McCombs School of Business 2023 – 2024

B.S. in Physics — The University of Texas at Austin 2016 – 2021